12.23 Stirling from Probability

WORKSHEET

The question

STEP II 2016, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If a discrete random variable X with mean μ and variance σ^2 is approximately normal, roughly what is $\mathrm{P}(X = \mu)$? (Think of the probability of an interval of width 1 centred at μ.)

STAGE 1 Sixteen coins

Covers: Part (i)

Ask yourself Find the mean and standard deviation of $\mathrm{B}(16, \frac{1}{2})$. Approximate $\mathrm{P}(X = 8)$ by the height of the normal density at its peak.

STAGE 2 The central binomial coefficient

Covers: Part (ii)

Pause and try For $\mathrm{B}(2n, \frac{1}{2})$, write $\mathrm{P}(X = n)$ exactly, and approximately by the same method. Set the two equal.

STAGE 3 Stirling's formula

Covers: Part (iii)

Ask yourself Do the same for a Poisson distribution with parameter n. What are its mean and variance? Write $\mathrm{P}(X = n)$ exactly and approximately.

What's really going on

How good is the approximation?

This argument is a heuristic: it uses the normal approximation *at a single point*, which the usual central limit theorem does not justify. A rigorous version, the *local central limit theorem*, does exactly that, and makes this a genuine proof. Stirling's formula is in fact remarkably accurate: the ratio $\frac{n!}{\sqrt{2\pi n}(n/e)^n}$ is about $1 + \frac{1}{12n}$, so even for $n = 1$ it gives 0.92 instead of 1, and for $n = 10$ it is 0.8% too small. The full series, $1 + \frac{1}{12n} + \frac{1}{288n^2} - \cdots$, is an example of an *asymptotic series*: it gives better and better approximations for large n, but for fixed n it eventually diverges.

Why factorials and normals are linked

The $\sqrt{2\pi}$ in Stirling's formula and in the normal density are the same constant, because $\int_{-\infty}^{\infty} e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}$. Abraham de Moivre found the normal approximation to the binomial in 1733 in exactly this way, needing an approximation to $\binom{n}{k}$ for large n; he found the form $n! \approx Cn^{n+1/2}e^{-n}$ but not the constant. James Stirling showed that $C = \sqrt{2\pi}$. The normal distribution was born from the asymptotics of factorials.

De Moivre and Stirling

Abraham de Moivre, a French Protestant who fled to London and earned a living tutoring and advising gamblers, published *The Doctrine of Chances* in 1718 and found the normal approximation to the binomial in a pamphlet of 1733, the first appearance of the normal curve. James Stirling, a Scottish mathematician, published the value of the constant in his *Methodus Differentialis* (1730), and de Moivre generously credited him. The approximation of factorials is still an active subject: computing $n!$ for huge n is needed in combinatorics and cryptography.

Try this next

1. **Check the formula.** Use Stirling's formula to estimate $10!$ and $20!$, and compare with the true values $3\,628\,800$ and $2\,432\,902\,008\,176\,640\,000$.

2. **From (iii) to (ii).** Show that Stirling's formula in part (iii) implies the approximation in part (ii).

3. **Digits of $100!$.** Use $\log_{10} n! \approx n \log_{10} \frac{n}{e} + \frac{1}{2}\log_{10}(2\pi n)$ to show that $100!$ has 158 digits.

4. **A tie in coin tossing.** Use part (ii) to show that the probability of exactly 500 heads in 1000 tosses of a fair coin is about 0.025.

Further reading

- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter II, which proves Stirling's formula.
- **Ronald Graham, Donald Knuth & Oren Patashnik, *Concrete Mathematics*** (Addison-Wesley), Chapter 9, on asymptotics.
- **Stephen Stigler, *The History of Statistics*** (Harvard University Press), on de Moivre and the birth of the normal curve.

The full guided solution is in Section 12.23:
https://beyondstep.keithmcnulty.org/ch12/stirling-probability.html